

JS + React 4-Day Study Plan

Theory Questions + Practical Tasks — Karthi's Revision Guide

Day 1 — JavaScript Foundations

Variables - Functions - Scope - Hoisting

Concepts: var / let / const | function declaration | arrow functions | scope | hoisting

Theory Questions

- Q1. var, let, const — difference enna? Oru example kodunga.
- Q2. Hoisting-la var vs let epdi differently behave aaaguthu? Explain with output.
- Q3. Arrow function vs normal function — this keyword epdi differ aaaguthu?
- Q4. Lexical scope-nu enna? Closure-oda connection enna?
- Q5. function greet(name = 'Karthi') {} — default parameter enna use case la useful?

Practical Task: Mini Calculator with Closures

makeCounter(start) function create pannunga — idu oru object return pannanum with 3 methods: increment(), decrement(), reset(). Counter value outside access aaagaama irukkanum (private-a). console.log la test pannunga.

Hint: Closure use pannunga — inner variable outer function scope la irukanum

[] Done

Day 2 — Arrays, Objects & Async

Array methods - Destructuring - Promises - Async/Await

Concepts: map / filter / reduce | destructuring | spread / rest | Promise | async/await

Theory Questions

- Q1. map() vs forEach() — enna difference? Enga use pannuvom?
- Q2. Promise states — pending, fulfilled, rejected — explain with real-life analogy.
- Q3. async/await vs .then().catch() — which is better and why?
- Q4. Spread operator vs Rest operator — same ... syntax, different use — explain.
- Q5. const arr = [1,2,3]; arr.push(4) — idu error varuma? Why or why not?

Practical Task: Student Grade Filter App

Oru students array create pannunga (name, marks). filter use pannu pass aaana students (marks >= 50) edungga. map use pannu each student-ku grade add pannunga (A/B/C). Finally reduce use pannu class average calculate pannunga. Console la output podunga.

Hint: Chain pannalaam — filter > map > reduce separately or together

[] Done

Day 3 — DOM Manipulation & Events

Selectors - Event Listeners - Form Validation - ES6 Modules

Concepts: querySelector | addEventListener | event delegation | preventDefault | template literals

Theory Questions

- Q1. getElementById vs querySelector — difference enna? Which one flexible?

Q2. Event bubbling enna? stopPropagation() enna pannum?

Q3. Event delegation — 100 buttons irundha, oru listener ■■■■■■? Explain.

Q4. innerHTML vs textContent — security angle la enna difference?

Q5. DOMContentLoaded vs window.onload — epdi differ aaaguthu?

Practical Task: Todo List with Vanilla JS

Pure HTML + JS la oru todo app build pannunga. Features: input box la type pann■ add button click-la item list-la appear aaaganum. Each item-la delete button irukanum. Empty input submit pannina validation message kaatanum. preventDefault() use pannunga.

Hint: DOM create pannunga dynamically — createElement, appendChild use pannunga

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Day 4 — React Kickstart

JSX - Components - Props - useState - useEffect

Concepts: JSX | functional components | props | useState | useEffect | Vite setup

Theory Questions

Q1. Virtual DOM enna? Real DOM la direct update pannaama React yen Virtual DOM use pannudu?

Q2. props vs state — difference enna? Oru analogy kodunga.

Q3. useEffect(callback, []) — empty dependency array enna meaning?

Q4. JSX la class use pannaama className use pannrom — why?

Q5. key prop in lists — yen mandatory? Index use pannradhu yen bad practice?

Practical Task: React Counter + Fetch App

Vite la React project create pannunga. 2 components: (1) Counter — increment, decrement, reset buttons with useState. (2) UserList — <https://jsonplaceholder.typicode.com/users> API call pann■ useEffect la, users list render pannunga (name + email). Loading state add pannunga.

Hint: fetch() or axios use pannalam. Loading: true initially, false after data arrives

[] Done

All the best Karthi! JS strong-a padichi React la fly pannuva! ■